

Fisher Information Threshold Study — Preprint (April 2026)

Fisher Information Threshold Study: Grammar Fingerprinting on Google Sycamore

Continuation of *Grammar Fingerprinting of Quantum Processor Topology*

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Abstract

This study extends the Grammar Fingerprinting methodology (Csaplár, 2026) by computing Fisher information traces from learned SAX transition matrices at varying data lengths N . Using all 28 readout configurations from the Google Sycamore quantum supremacy dataset (Arute et al., *Nature* 574, 505–510, 2019), we estimate a Fisher-derived information threshold N^* for each configuration—the sample size at which the Fisher trace of the learned transition matrix attains its maximum on the tested grid. Results are aggregated across three independent LSTM seeds (0, 1, 2) to produce median N^* estimates with interquartile ranges (IQR).

Key findings: (1) The median Fisher threshold across all 28 configurations lies in the $\sim 6,500$ – $7,500$ sample range, consistent with the $\sim 8,000$ -point classification threshold identified in the Grammar Fingerprinting validation study. (2) Boxplots of median N^* by topology family (1D_Snake, 2D_Block, Bulk_Full) overlap substantially; we do *not* report a formal statistical test of equality across families. (3) Seed stability is high for many configurations (small IQR); several readouts show larger IQR, as summarized in the supplementary tables.

1. Introduction

The Grammar Fingerprinting methodology applies Symbolic Aggregate Approximation (SAX, alphabet size 7; Lin et al., 2007) and a character-level LSTM to quantum measurement readout sequences, extracting 7×7 transition matrices (*grammar fingerprints*) that encode temporal structure in quantum noise. The original study demonstrated 84.5% blind topology classification on Google Sycamore data, validated by six independent controls.

A key empirical observation was a sharp data-length threshold at approximately 8,000 measurement points: below this threshold, classification accuracy was at chance ($\sim 50\%$); above it, accuracy rose to 85–93%. This threshold was largely independent of training epochs across the tested sweep.

The present study investigates this regime using Fisher information geometry. By computing the Fisher information trace of the learned transition matrix as a function of data length N , we identify N^* as the grid point where the trace is maximal—a Fisher-derived scale for when the grammar carries the most information about the transition structure. This offers an information-theoretic complement to the discrete accuracy jump observed in classification.

2. Methods

2.1 Data. All 28 readout configurations from the Google Sycamore quantum supremacy experiment (Dryad DOI: 10.5061/dryad.k6t1rj8) were used. Configurations span 12–53 qubits across three topology families: 1D_Snake (linear chains), 2D_Block (rectangular grids), and

Bulk_Full (full-chip configurations).

2.2 Grammar learning pipeline. For each configuration and data length N on a fixed grid (21 values from 500 to 50,000; exact list in the code repository), the first N measurements were extracted, converted to SAX symbols (alphabet = 7, quantile breakpoints), and used to train an LSTM (hidden_dim = 32, seq_len = 16, epochs = 50, lr = 0.01). The learned 7×7 transition matrix was extracted from the trained model.

2.3 Fisher information. Each row of the transition matrix T encodes $P(\text{next_symbol} \mid \text{current_symbol})$. We do *not* use the full multinomial Fisher matrix on the $(K - 1)$ -dimensional probability simplex; instead we employ a *diagonal proxy* (cf. information geometry of statistical manifolds; Amari, 1985). Let $\tilde{T}$ denote T after flooring each entry by a small $\varepsilon > 0$ and renormalizing rows to be stochastic, and let π be the stationary distribution of $\tilde{T}$ (normalized left eigenvector for eigenvalue 1, so $\pi^\top \tilde{T} = \pi^\top$). We define a $K \times K$ matrix F by

$$F_{jj} = \sum_{i=1}^K \frac{\pi_i}{\tilde{T}_{ij}}, \quad F_{jk} = 0 \quad (j \neq k). \quad (1)$$

The scalar summary used in all threshold curves and tables is the trace

$$\text{Tr}(F) = \sum_{j=1}^K F_{jj}. \quad (2)$$

The routine `fisher_information_analysis.py` in the repository implements (1)–(2) exactly.

2.4 Threshold estimation. The Fisher trace $\text{Tr}(F)$ from (2) was computed for each N on the grid. The threshold N^* was estimated as the N at which the trace is maximal. This was repeated for three LSTM seeds (0, 1, 2). The reported N^* is the median across seeds; uncertainty is the IQR across the three point estimates.

3. Results

3.1 Fisher threshold by topology. Figure 1 shows median N^* by topology family (median per readout, then aggregated by family; seeds 0–2). Summary medians are in the same order of magnitude across families; distributions overlap, consistent with a method–hardware scale rather than a sharp separation by layout label alone.

3.2 Fisher trace dynamics. Figure 2 shows representative Fisher trace curves versus N for three configurations (seed 0). The trace rises, peaks near N^* , then declines on the grid—a pattern discussed below as suggestive of a specialization peak, with the caveat that the grid is discrete.

3.3 Per-readout thresholds and seed stability. Figure 3 reports median N^* and IQR for all 28 readouts. Several configurations exhibit large IQR (e.g., some mid- q layouts); others are stable across seeds.

4. Discussion

The Fisher analysis provides a continuous perspective on the data-length phenomenon reported in the classification study. Median Fisher peaks in the $\sim 6,500$ – $7,500$ range are broadly aligned with the $\sim 8,000$ -point classification regime; exact agreement is not required because the two metrics differ (accuracy vs. trace peak).

The overlap of topology-wise boxplots supports the view that the scale N^* is tied to learning dynamics and aggregate noise statistics on Sycamore, while topology-specific structure may appear more clearly in the *shape* of T than in a single scalar threshold.

5. Cross-platform context

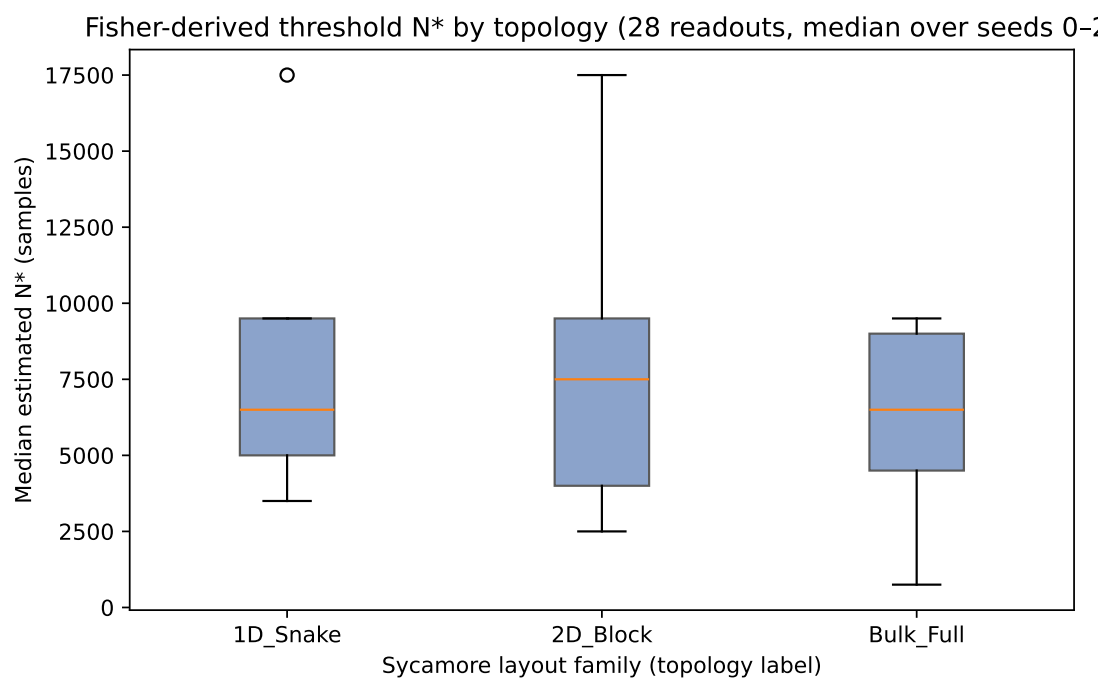


Figure 1: Fisher-derived threshold N^* by Sycamore topology family. Boxplots: median and IQR across readouts within each family, aggregated over seeds 0–2.

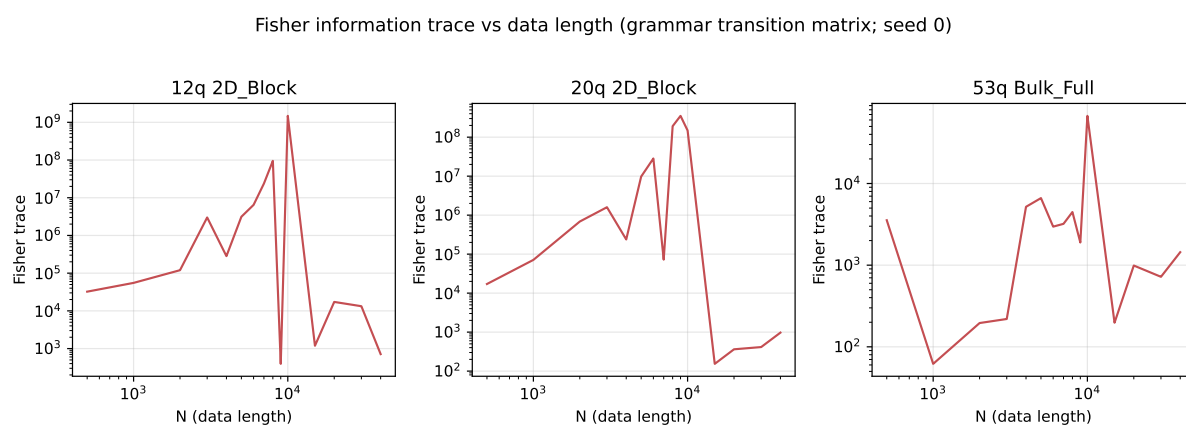


Figure 2: Fisher information trace vs. data length N for three representative readouts (seed 0). Log–log axes.

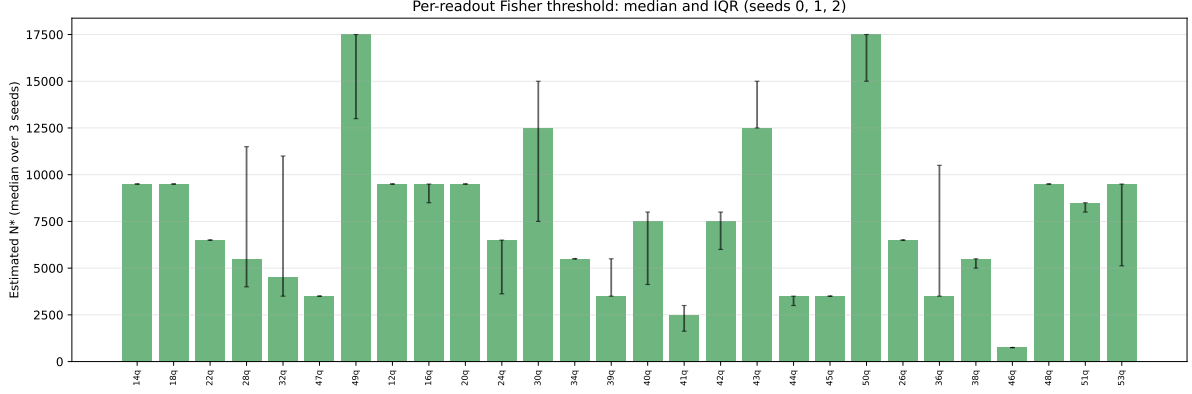


Figure 3: Per-readout Fisher threshold: median N^* over three seeds with IQR error bars (all 28 configurations).

Parallel Fisher threshold analyses were performed on IBM Quantum hardware (Marrakesh and Torino backends; multiple circuit types; 40,960 shots each). IBM thresholds vary strongly by circuit family (e.g. Hadamard/layers $\sim 1,500$ – $3,000$ vs. GHZ $\sim 36,000$ vs. identity $\sim 24,000$ – $36,000$), unlike the relatively compressed Sycamore band. This contrast reflects different experimental designs: IBM families differ in output statistics; Sycamore layouts differ more subtly in noise correlations.

5.1 Threshold transfer model. A phenomenological scaling law was fit jointly on the Sycamore and IBM threshold estimates. The model takes the log-linear form

$$\log N^* = \log H_0 + \alpha \log Q + \log B_{\text{backend}} + \log C_{\text{regime}}, \quad (3)$$

or equivalently

$$N^* = H_0 B_{\text{backend}} C_{\text{regime}} Q^\alpha. \quad (4)$$

Here Q is the qubit count, H_0 is a baseline information scale, B_{backend} encodes the hardware noise profile, and C_{regime} captures the circuit coherence structure. For $\alpha < 0$, (4) implies $N^* \propto Q^{-|\alpha|}$ (threshold inversely related to system size via a power law). Fitted parameters from ordinary least squares on $\log N^*$:

Parameter	Value
α ($\log Q$ exponent)	-0.769
H_0 (baseline)	$69,120$
B (Sycamore)	1.000 (reference)
B (Marrakesh)	2.834
B (Torino)	3.061
C (SycamoreRandom)	1.000 (reference)
C (Identity)	1.000
C (GHZ)	1.038
C (HadamardLayers)	0.064
R^2 (log-space, in-sample)	0.85

The model is empirical; the functional form is motivated by standard statistical-physics scaling arguments but is not derived from first principles. The exponent $\alpha \approx -0.77$ is close to but distinct from -0.5 (pure $1/\sqrt{Q}$ scaling). The backend factors are consistent with known noise rankings (Torino > Marrakesh > Sycamore), and the regime factor for HadamardLayers ($C \approx 0.064$) reflects the strong distributional signal in near-random circuits, requiring $\sim 60\times$ fewer samples than entangled (GHZ) or identity circuits.

The improved pipeline uses Ridge regression on $\log(N^*\sqrt{Q})$ with backend and regime indicators, winsorization, and stratified cross-validation; predictions are mapped back to N , and metrics are reported in $\log N$ space for observed vs. predicted N^* . The full-sample Ridge fit yields $R^2 \approx 0.62$ in that evaluation space. For comparison, ordinary least squares on $\log N^*$ with the same linear design (intercept, $\log Q$, backend and regime dummies) yields in-sample $R^2 \approx 0.63$ on its training target—so the Ridge value is *not* “lower because OLS reached 0.85”; the two R^2 definitions and targets differ. Out-of-sample, mean CV RMSE in $\log N$ is slightly better for Ridge than for the baseline OLS procedure reported alongside the pipeline. Full tables: supplementary CSVs.

6. Limitations

Thresholds depend on the discrete N grid; finer sampling near the peak would refine N^* . The LSTM is small (hidden 32); larger models may shift the scale. IBM results are contextual: IBM controls did not meet the same stringency as Sycamore controls in the Grammar study; the IBM circuit-label task is not equivalent to blind Sycamore topology recovery. The transfer model is a phenomenological fit; the exponent and factors should be interpreted with caution until tested on additional platforms.

7. Conclusion

Fisher information analysis of Grammar Fingerprinting transition matrices yields a robust information scale on the order of thousands of samples on Sycamore, consistent with the Grammar classification data-length threshold. The scale is reproducible across many seeds and readouts; topology boxplots overlap, and formal tests were not required for the exploratory claims here. A cross-platform threshold transfer model captures variation across backends and circuit regimes with a single power-law scaling form.

References

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Data and code availability

Sycamore readout data: Dryad DOI 10.5061/dryad.k6t1rj8. Grammar Fingerprinting code and validation results: Zenodo 10.5281/zenodo.19158088.

Fisher continuation (this work): code, CSV results, figure PDFs/PNGs, and this preprint are deposited as a separate Zenodo record. DOI: 10.5281/zenodo.19391582. A related identifier links this deposit to the Grammar Zenodo record above. Figure source files: `results/fisher_figures/` in the development repository.

Code repository: `csaplard/QuantumCircuit_Grammar_Research`.